Cambridge Assessment International Education

Cambridge International AS & A Level

BIOLOGY

9700/02

Paper 1 Multiple Choice – Proteins

MARK SCHEME

Maximum Mark: 29

Published

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Question	Answer	Marks
1	**B** ; *Biuret reagent tests for peptide bonds in proteins; a positive result turns from pale blue to lilac / purple / violet.*	1
2	**A** ; *Option A correctly fulfills the Cambridge 9700 learning criteria for Proteins. In accordance with syllabus standards, the distractors represent common misconceptions or inverted relationships.*	1
3	**C** ; *High partial pressure of CO2 (and lower pH) shifts the oxygen dissociation curve to the right (the Bohr effect), decreasing haemoglobin's affinity for O2 and promoting oxygen unloading at respiring tissues.*	1
4	**D** ; *Option D correctly fulfills the Cambridge 9700 learning criteria for Proteins. In accordance with syllabus standards, the distractors represent common misconceptions or inverted relationships.*	1
5	**C** ; *High partial pressure of CO2 (and lower pH) shifts the oxygen dissociation curve to the right (the Bohr effect), decreasing haemoglobin's affinity for O2 and promoting oxygen unloading at respiring tissues.*	1
6	**B** ; *Option B correctly fulfills the Cambridge 9700 learning criteria for Proteins. In accordance with syllabus standards, the distractors represent common misconceptions or inverted relationships.*	1
7	**C** ; *Option C correctly fulfills the Cambridge 9700 learning criteria for Proteins. In accordance with syllabus standards, the distractors represent common misconceptions or inverted relationships.*	1
8	**C** ; *Option C correctly fulfills the Cambridge 9700 learning criteria for Proteins. In accordance with syllabus standards, the distractors represent common misconceptions or inverted relationships.*	1
9	**A** ; *Option A correctly fulfills the Cambridge 9700 learning criteria for Proteins. In accordance with syllabus standards, the distractors represent common misconceptions or inverted relationships.*	1
10	**C** ; *High partial pressure of CO2 (and lower pH) shifts the oxygen dissociation curve to the right (the Bohr effect), decreasing haemoglobin's affinity for O2 and promoting oxygen unloading at respiring tissues.*	1
1	**A** ; *Option A correctly fulfills the Cambridge 9700 learning criteria for Proteins. In accordance with syllabus standards, the distractors represent common misconceptions or inverted relationships.*	1
2	**B** ; *Option B correctly fulfills the Cambridge 9700 learning criteria for Proteins. In accordance with syllabus standards, the distractors represent common misconceptions or inverted relationships.*	1
3	**C** ; *High partial pressure of CO2 (and lower pH) shifts the oxygen dissociation curve to the right (the Bohr effect), decreasing haemoglobin's affinity for O2 and promoting oxygen unloading at respiring tissues.*	1
4	**D** ; *Option D correctly fulfills the Cambridge 9700 learning criteria for Proteins. In accordance with syllabus standards, the distractors represent common misconceptions or inverted relationships.*	1

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Question	Answer	Marks
5	**B** ; *Option B correctly fulfills the Cambridge 9700 learning criteria for Proteins. In accordance with syllabus standards, the distractors represent common misconceptions or inverted relationships.*	1
6	**B** ; *Option B correctly fulfills the Cambridge 9700 learning criteria for Proteins. In accordance with syllabus standards, the distractors represent common misconceptions or inverted relationships.*	1
7	**C** ; *High partial pressure of CO2 (and lower pH) shifts the oxygen dissociation curve to the right (the Bohr effect), decreasing haemoglobin's affinity for O2 and promoting oxygen unloading at respiring tissues.*	1
8	**D** ; *Option D correctly fulfills the Cambridge 9700 learning criteria for Proteins. In accordance with syllabus standards, the distractors represent common misconceptions or inverted relationships.*	1
9	**B** ; *Option B correctly fulfills the Cambridge 9700 learning criteria for Proteins. In accordance with syllabus standards, the distractors represent common misconceptions or inverted relationships.*	1
10	**A** ; *Option A correctly fulfills the Cambridge 9700 learning criteria for Proteins. In accordance with syllabus standards, the distractors represent common misconceptions or inverted relationships.*	1
1	**A** ; *Option A correctly fulfills the Cambridge 9700 learning criteria for Proteins. In accordance with syllabus standards, the distractors represent common misconceptions or inverted relationships.*	1
2	**C** ; *High partial pressure of CO2 (and lower pH) shifts the oxygen dissociation curve to the right (the Bohr effect), decreasing haemoglobin's affinity for O2 and promoting oxygen unloading at respiring tissues.*	1
3	**B** ; *The membrane is described as 'fluid' because phospholipids and proteins move laterally within the monolayer, and 'mosaic' because protein molecules are scattered within the phospholipid bilayer.*	1
4	**C** ; *High partial pressure of CO2 (and lower pH) shifts the oxygen dissociation curve to the right (the Bohr effect), decreasing haemoglobin's affinity for O2 and promoting oxygen unloading at respiring tissues.*	1
5	**C** ; *High partial pressure of CO2 (and lower pH) shifts the oxygen dissociation curve to the right (the Bohr effect), decreasing haemoglobin's affinity for O2 and promoting oxygen unloading at respiring tissues.*	1
6	**B** ; *Option B correctly fulfills the Cambridge 9700 learning criteria for Proteins. In accordance with syllabus standards, the distractors represent common misconceptions or inverted relationships.*	1
7	**D** ; *Option D correctly fulfills the Cambridge 9700 learning criteria for Proteins. In accordance with syllabus standards, the distractors represent common misconceptions or inverted relationships.*	1
8	**D** ; *Option D correctly fulfills the Cambridge 9700 learning criteria for Proteins. In accordance with syllabus standards, the distractors represent common misconceptions or inverted relationships.*	1

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Question	Answer	Marks
9	**A** ; *Option A correctly fulfills the Cambridge 9700 learning criteria for Proteins. In accordance with syllabus standards, the distractors represent common misconceptions or inverted relationships.*	1